

1. Entity Identification

- AIRPORT — airport
- AIRLINE — airline
- FLIGHT — flight
- PASSENGER — passenger
- BOOKING — booking
- BOOKING_CHANGE — booking change history
- BOARDING_PASS — boarding pass
- BAGGAGE — baggage
- BAGGAGE_CHECK — baggage check
- SECURITY_CHECK — security check

AIRPORT

Attribute	Data Type	Key	Constraints
airport_id	INT	PK	
airport_name	VARCHAR		NOT NULL
Country	VARCHAR		NOT NULL
State	VARCHAR		NOT NULL
City	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

AIRLINE

Attribute	Data Type	Key	Constraints
airline_id	INT	PK	
airline_code	VARCHAR		UNIQUE, NOT NULL
Name	VARCHAR		NOT NULL
Country	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

FLIGHT

Attribute	Data Type	Key	Constraints
flight_id	INT	PK	
airline_id	INT	FK -> AIRLINE.airline_id	NOT NULL
departure_airport_id	INT	FK -> AIRPORT.airport_id	NOT NULL
arrival_airport_id	INT	FK -> AIRPORT.airport_id	NOT NULL
departing_gate	VARCHAR		
arriving_gate	VARCHAR		
scheduled_departure_time	DATETIME		NOT NULL
scheduled_arrival_time	DATETIME		NOT NULL
actual_departure_time	DATETIME		
actual_arrival_time	DATETIME		
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

PASSENGER

Attribute	Data Type	Key	Constraints
passenger_id	INT	PK	
first_name	VARCHAR		NOT NULL
last_name	VARCHAR		NOT NULL
gender	VARCHAR		
date_of_birth	DATE		NOT NULL
country_of_citizenship	VARCHAR		NOT NULL
country_of_residence	VARCHAR		
passport_number	VARCHAR		UNIQUE, NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BOOKING

Attribute	Data Type	Key	Constraints
booking_id	INT	PK	
flight_id	INT	FK -> FLIGHT.flight_id	NOT NULL
passenger_id	INT	FK -> PASSENGER.passenger_id	NOT NULL
status	VARCHAR		NOT NULL
booking_platform	VARCHAR		
ticket_price	DECIMAL(10,2)		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BOOKING_CHANGE

Attribute	Data Type	Key	Constraints
booking_change_id	INT	PK	
booking_id	INT	FK -> BOOKING.booking_id	NOT NULL
change_description	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BOARDING_PASS

Attribute	Data Type	Key	Constraints
boarding_pass_id	INT	PK	
booking_id	INT	FK -> BOOKING.booking_id	NOT NULL
seat	VARCHAR		NOT NULL
boarding_time	DATETIME		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BAGGAGE

Attribute	Data Type	Key	Constraints
baggage_id	INT	PK	
booking_id	INT	FK -> BOOKING.booking_id	NOT NULL
weight_kg	DECIMAL(6,2)		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

BAGGAGE_CHECK

Attribute	Data Type	Key	Constraints
baggage_checking_id	INT	PK	
booking_id	INT	FK -> BOOKING.booking_id	NOT NULL
passenger_id	INT	FK -> PASSENGER.passenger_id	NOT NULL
check_results	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

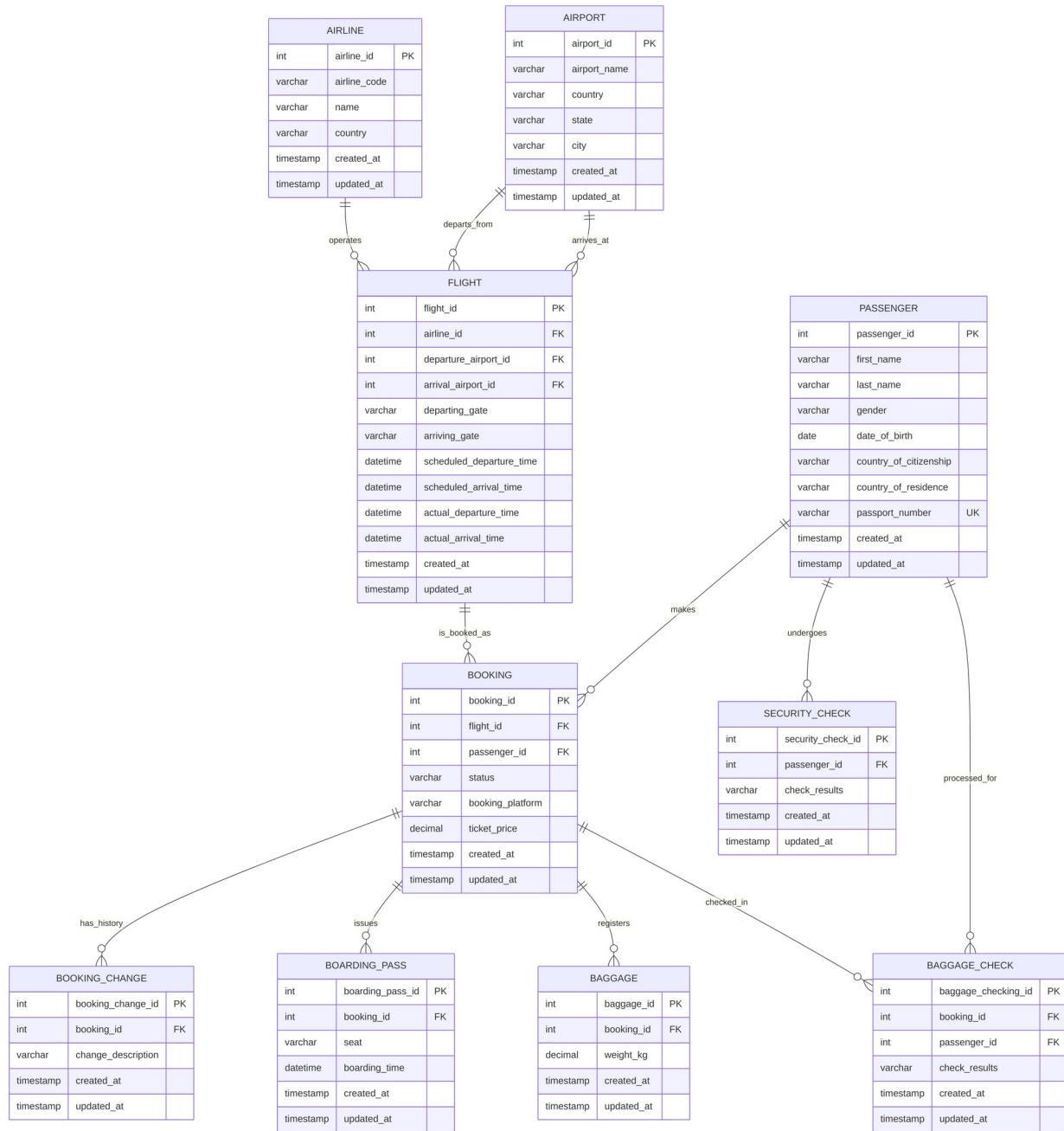
SECURITY_CHECK

Attribute	Data Type	Key	Constraints
security_check_id	INT	PK	
passenger_id	INT	FK -> PASSENGER.passenger_id	NOT NULL
check_results	VARCHAR		NOT NULL
created_at	TIMESTAMP		NOT NULL
updated_at	TIMESTAMP		NOT NULL

3. Relationships and Cardinality

Entity A	Entity B	Cardinality
AIRLINE	FLIGHT	1 : N
AIRPORT	FLIGHT (departure)	1 : N
AIRPORT	FLIGHT (arrival)	1 : N
FLIGHT	BOOKING	1 : N
PASSENGER	BOOKING	1 : N
BOOKING	BOOKING_CHANGE	1 : N
BOOKING	BOARDING_PASS	1 : N
BOOKING	BAGGAGE	1 : N
BOOKING	BAGGAGE_CHECK	1 : N
PASSENGER	BAGGAGE_CHECK	1 : N
PASSENGER	SECURITY_CHECK	1 : N

4–6. ER Diagram



7. Legend

- PK = primary key (unique identifier of a record).
- FK = foreign key (reference to the primary key of another entity).
- UK = alternate unique key (for example, passport_number).
- The “||” symbol at the end of a line = “exactly one” (mandatory participation).
- The “o{” symbol at the end of a line = “zero or many” (optional, multiple participation).